

EEE 4238: Biomedical Instrumentation Lab

Experiment 5: ECG Analysis with MATLAB

Group 2
Section C2

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1. Objective

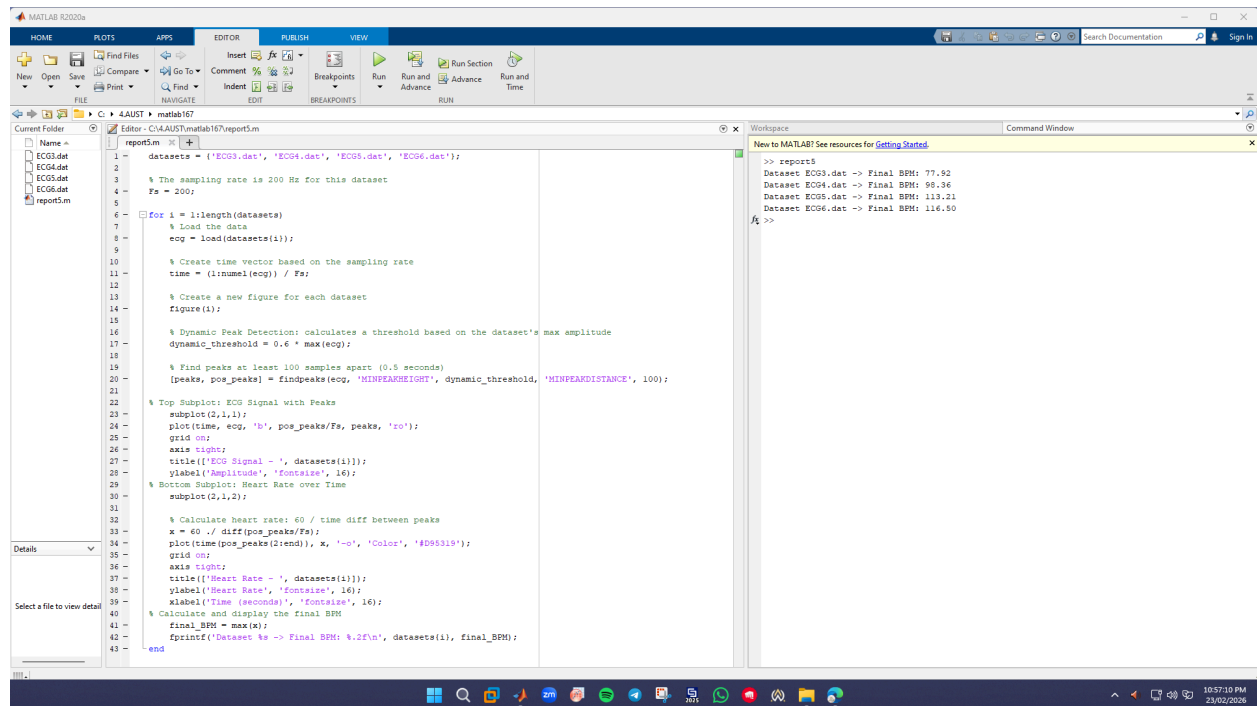
To load, process, and display electrocardiogram (ECG) data using MATLAB, detect the R-peaks within the signal, and calculate the inferred heart rate in Beats Per Minute (BPM) from the time differences between these peaks.

2. Methodology

You can plug in the paragraph we wrote earlier here: For this lab experiment, MATLAB was utilized to process and analyze four separate electrocardiogram (ECG) datasets. Each `.dat` file was loaded into the workspace, and a corresponding time vector was generated based on the dataset's sampling frequency of 200 Hz. To accurately identify the R-peaks within the ECG waveform, MATLAB's built-in `findpeaks` function was employed.

A dynamic minimum peak height was implemented alongside a minimum peak distance to reliably isolate the peaks and filter out noise. The instantaneous heart rate was then inferred by determining the time differences between consecutive R-peaks using the `diff` function, dividing 60 by those intervals to convert the data into beats per minute (BPM). Finally, the maximum value of these calculated rates was extracted to determine the final BPM for each respective dataset.

3. MATLAB Code



The image shows the MATLAB R2020a interface with a script editor, workspace, and command window. The script, named 'report5.m', processes four ECG datasets (ECG3.dat, ECG4.dat, ECG5.dat, ECG6.dat) to calculate heart rate (BPM) and generate a report.

```
1 datasets = {'ECG3.dat', 'ECG4.dat', 'ECG5.dat', 'ECG6.dat'};
2
3 % The sampling rate is 200 Hz for this dataset
4 Fs = 200;
5
6 for i = 1:length(datasets)
7     % Load the data
8     ecg = load(datasets{i});
9
10    % Create time vector based on the sampling rate
11    time = (1:numel(ecg)) / Fs;
12
13    % Create a new figure for each dataset
14    figure(i);
15
16    % Dynamic Peak Detection: calculates a threshold based on the dataset's max amplitude
17    dynamic_threshold = 0.6 * max(ecg);
18
19    % Find peaks at least 100 samples apart (0.5 seconds)
20    [peaks, pos_peaks] = findpeaks(ecg, 'MINPEAKHEIGHT', dynamic_threshold, 'MINPEAKDISTANCE', 100);
21
22    % Top Subplot: ECG Signal with Peaks
23    subplot(2,1,1);
24    plot(time, ecg, 'b', pos_peaks/Fs, peaks, 'ro');
25    grid on;
26    axis tight;
27    title(['ECG Signal - ', datasets{i}]);
28    ylabel('Amplitude', 'fontsize', 16);
29    % Bottom Subplot: Heart Rate over Time
30    subplot(2,1,2);
31
32    % Calculate heart rate: 60 / time diff between peaks
33    x = 60 ./ diff(pos_peaks/Fs);
34    plot(time(pos_peaks(2:end)), x, '-o', 'Color', '#D95319');
35    grid on;
36    axis tight;
37    title(['Heart Rate - ', datasets{i}]);
38    ylabel('Heart Rate', 'fontsize', 16);
39    xlabel('Time (seconds)', 'fontsize', 16);
40    % Calculate and display the final BPM
41    final_BPM = max(x);
42    fprintf('Dataset %s -> Final BPM: %.2f\n', datasets{i}, final_BPM);
43 end
```

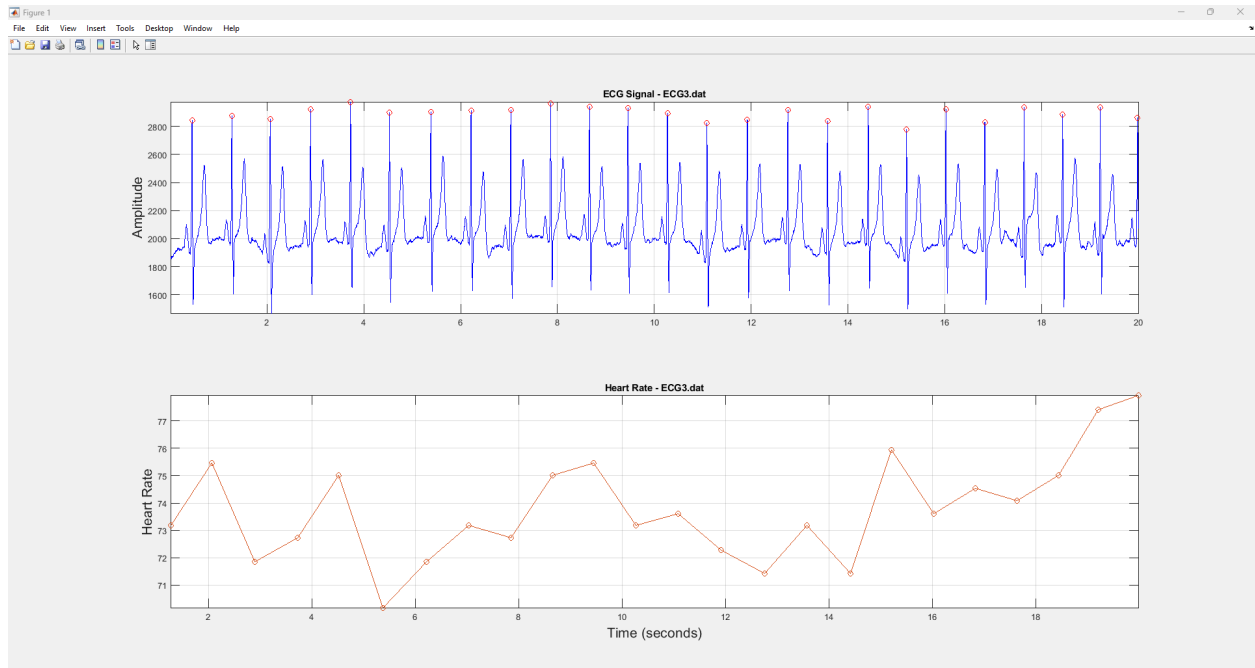
The workspace shows the results of the script execution:

```
>> report5
Dataset ECG3.dat -> Final BPM: 77.90
Dataset ECG4.dat -> Final BPM: 99.36
Dataset ECG5.dat -> Final BPM: 119.21
Dataset ECG6.dat -> Final BPM: 116.50
```

The command window shows the output of the script, displaying the final BPM for each dataset.

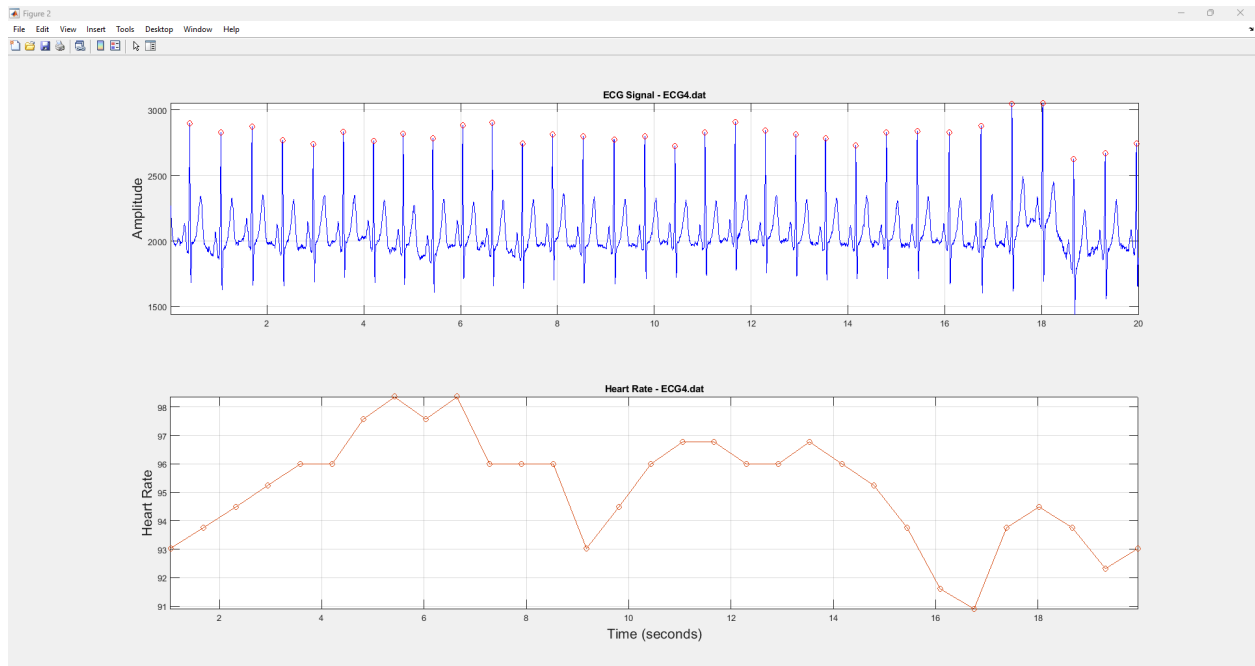
4. Results and Analysis

4.1 Dataset: ECG3.dat



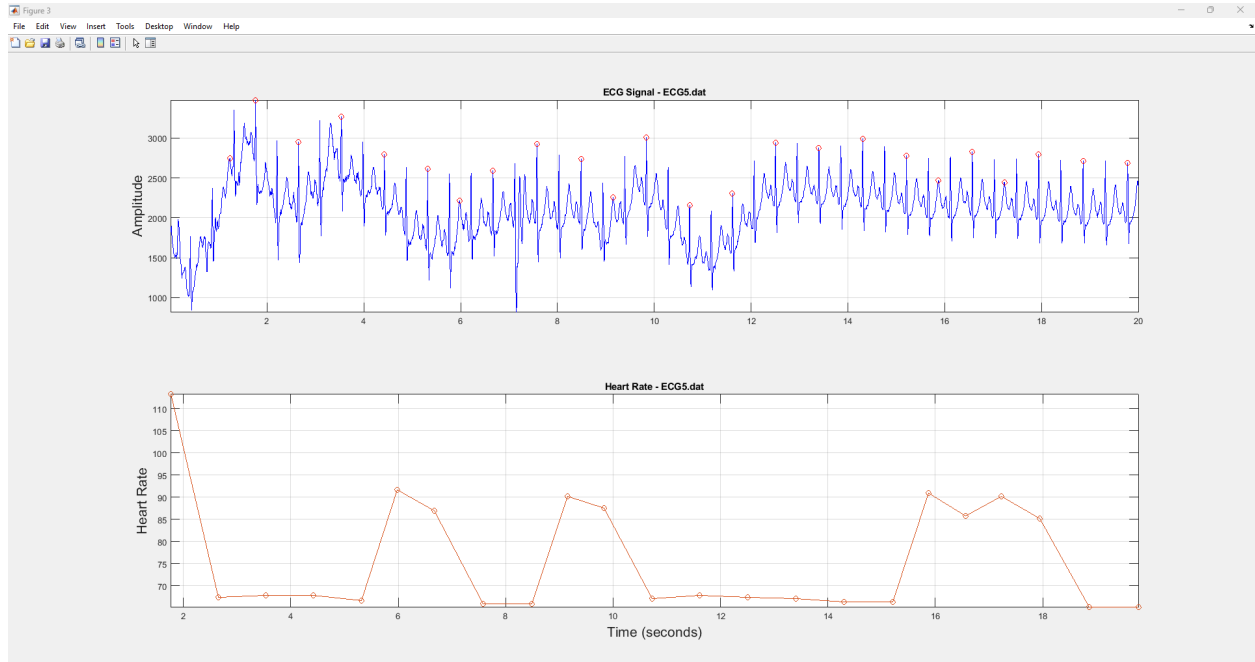
Final BPM: 78

4.2 Dataset: ECG4.dat



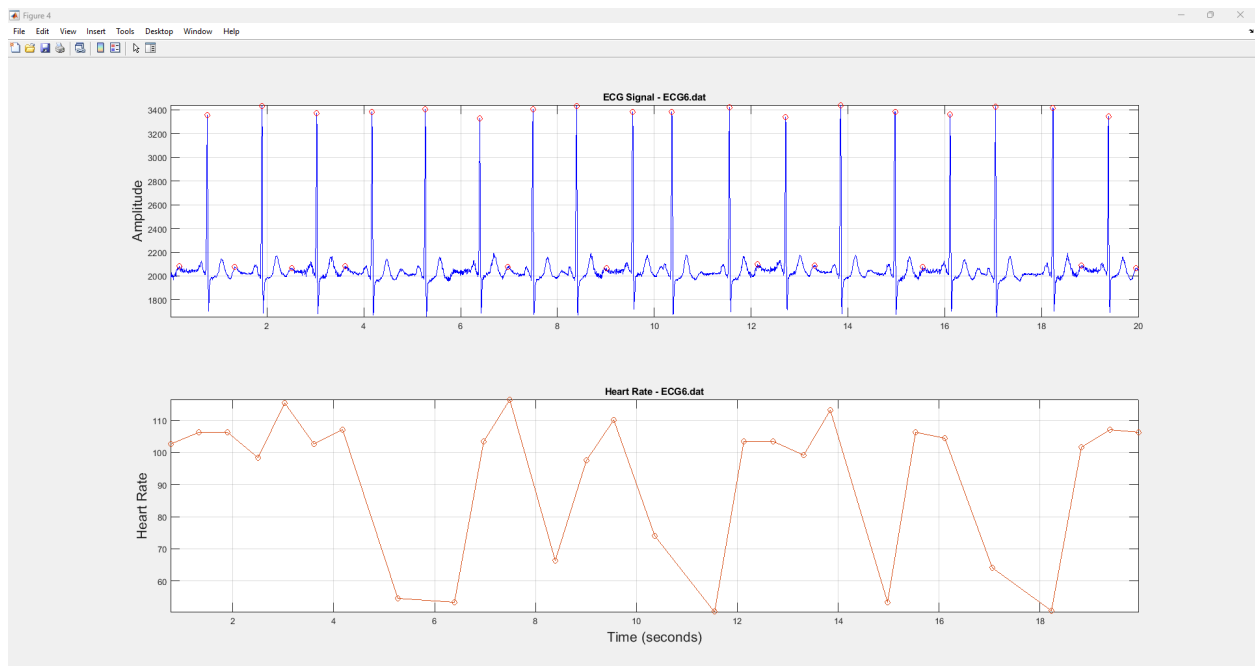
Final BPM: 98

4.3 Dataset: ECG5.dat



Final BPM: 113

4.4 Dataset: ECG6.dat



Final BPM: 116

Discussion & Conclusion

he experiment successfully demonstrated the relationship between the R-R interval of an ECG signal and the resulting heart rate. By isolating the sharp R-peaks, which represent the depolarization of the ventricles, we were able to programmatically calculate the time between consecutive QRS complexes and output the final BPM for four distinct patient datasets.

Furthermore, analyzing these final BPM values allows us to classify the physiological state of the heart. A normal adult heart rate falls between 60 and 100 bpm. Rates exceeding 100 bpm indicate tachycardia , while rates dropping below 60 bpm indicate bradycardia.

ECG3.dat: The final BPM was 77.92, which indicates a normal heart rate.

ECG4.dat: The final BPM was 98.36, which indicates a normal heart rate (though nearing the upper limit of the normal resting range).

ECG5.dat: The final BPM was 113.21, which indicates tachycardia.

ECG6.dat: The final BPM was 116.50, which also indicates tachycardia.